

Deployment Guide

Prerequisites

- Node.js 18+ installed
- pnpm package manager
- Expo CLI (`npm install -g expo-cli`)
- EAS CLI (`npm install -g eas-cli`)
- Git configured
- AWS S3 credentials (for storage)

Environment Setup

1. Create `.env.local` file

Bash

```
# Database
DATABASE_URL=postgresql://user:password@localhost:5432/social_chat

# JWT
JWT_SECRET=your-super-secret-jwt-key-here

# OAuth
GOOGLE_CLIENT_ID=your-google-client-id
GOOGLE_CLIENT_SECRET=your-google-client-secret

# S3 Storage
AWS_ACCESS_KEY_ID=your-aws-key
AWS_SECRET_ACCESS_KEY=your-aws-secret
AWS_S3_BUCKET=your-bucket-name
AWS_REGION=us-east-1

# Server
PORT=3000
NODE_ENV=production
API_URL=https://your-api-domain.com
```

Building APK (Android)

Option 1: Using EAS Build (Recommended)

Bash

```
# Login to Expo  
eas login  
  
# Configure EAS  
eas build:configure  
  
# Build APK  
eas build --platform android --type apk  
  
# Download APK  
# The APK will be available at the provided URL
```

Option 2: Local Build

Bash

```
# Install dependencies  
pnpm install  
  
# Build APK locally  
expo build:android -t apk  
  
# This requires Android SDK and build tools installed
```

Building IPA (iOS)

Bash

```
# Build IPA  
eas build --platform ios  
  
# Download IPA  
# The IPA will be available at the provided URL
```

Web Deployment

Deploy to Cloud Run

Bash

```
# Build the app
pnpm run build

# Create Dockerfile
cat > Dockerfile << 'EOF'
FROM node:18-alpine

WORKDIR /app

COPY package*.json ./
RUN npm ci --only=production

COPY . .

EXPOSE 3000

CMD ["npm", "start"]
EOF

# Build Docker image
docker build -t social-chat-app .

# Push to Google Container Registry
gcloud builds submit --tag gcr.io/PROJECT_ID/social-chat-app

# Deploy to Cloud Run
gcloud run deploy social-chat-app \
  --image gcr.io/PROJECT_ID/social-chat-app \
  --platform managed \
  --region us-central1 \
  --allow-unauthenticated
```

Deploy to Vercel

Bash

```
# Install Vercel CLI
npm i -g vercel

# Deploy
vercel

# Follow the prompts
```

Database Migration

Run Migrations

Bash

```
# Generate migrations
pnpm run db:push

# Migrate database
pnpm run db:migrate
```

Seed Database (Optional)

Bash

```
# Create seed script in scripts/seed.ts
pnpm run db:seed
```

Post-Deployment Checklist

- ☐ Verify environment variables are set
- ☐ Run database migrations
- ☐ Test authentication flow
- ☐ Verify S3 storage connectivity
- ☐ Test messaging system
- ☐ Test calls functionality
- ☐ Monitor error logs
- ☐ Set up monitoring and alerts
- ☐ Configure CDN (if applicable)
- ☐ Enable HTTPS/SSL
- ☐ Set up backup strategy
- ☐ Configure rate limiting
- ☐ Test admin dashboard
- ☐ Verify push notifications

- ☐ Test cross-platform compatibility

Monitoring & Logging

Set Up Monitoring

Bash

```
# View logs
gcloud logging read "resource.type=cloud_run_revision AND resource.labels.service"

# Set up alerts
gcloud alpha monitoring policies create \
  --notification-channels=CHANNEL_ID \
  --display-name="Social Chat App Errors"
```

Error Tracking

- Integrate Sentry for error tracking
- Set up log aggregation (Stackdriver, DataDog)
- Configure performance monitoring

Scaling

Horizontal Scaling

Bash

```
# Increase Cloud Run instances
gcloud run services update social-chat-app \
  --max-instances 100 \
  --min-instances 5
```

Database Scaling

- Use read replicas for high traffic
- Implement connection pooling
- Optimize queries and add indexes
- Consider sharding for very large datasets

Backup & Recovery

Database Backup

Bash

```
# Backup PostgreSQL  
pg_dump -h localhost -U user -d social_chat > backup.sql  
  
# Restore from backup  
psql -h localhost -U user -d social_chat < backup.sql
```

S3 Backup

Bash

```
# Enable S3 versioning  
aws s3api put-bucket-versioning \  
  --bucket your-bucket \  
  --versioning-configuration Status=Enabled  
  
# Enable S3 replication for disaster recovery
```

Security Hardening

SSL/TLS

Bash

```
# Use Let's Encrypt for free SSL  
certbot certonly --standalone -d your-domain.com
```

Firewall Rules

Bash

```
# Restrict access to database  
gcloud compute firewall-rules create restrict-db \  
  --allow=tcp:5432 \  
  --source-ranges=CLOUD_RUN_IP
```

API Rate Limiting

- Implement rate limiting per user
- Set up DDoS protection
- Use API keys for third-party access

Troubleshooting

Common Issues

APK Build Fails

- Clear cache: `expo build:android --clear`
- Check Java version: `java -version`
- Verify Android SDK: `echo $ANDROID_HOME`

Database Connection Error

- Verify DATABASE_URL format
- Check network connectivity
- Ensure database is running

S3 Upload Fails

- Verify AWS credentials
- Check S3 bucket permissions
- Ensure bucket exists in correct region

Socket.io Connection Issues

- Check firewall rules
- Verify WebSocket support
- Enable CORS for your domain

Performance Optimization

Frontend

- Enable code splitting
- Implement lazy loading
- Optimize images

- Minify CSS/JS
- Use CDN for static assets

Backend

- Add database indexes
- Implement caching (Redis)
- Use connection pooling
- Optimize queries
- Enable gzip compression

Monitoring

Bash

```
# Monitor performance
gcloud monitoring dashboards create \
  --config-from-file=dashboard.yaml
```

Rollback Procedure

Bash

```
# If deployment fails, rollback to previous version
gcloud run deploy social-chat-app \
  --image gcr.io/PROJECT_ID/social-chat-app:PREVIOUS_TAG \
  --platform managed \
  --region us-central1
```

Support

For deployment issues, check:

- Expo documentation: <https://docs.expo.dev>
- Cloud Run documentation: <https://cloud.google.com/run/docs>
- PostgreSQL documentation: <https://www.postgresql.org/docs>
- Socket.io documentation: <https://socket.io/docs>